

Chat App

A real-time chat application built with Node.js, Socket.io, and vanilla JavaScript. Features multiple chat rooms, private messaging, emoji reactions, and much more — all with zero login required.

Features

-  **Real-time messaging** — messages appear instantly for everyone in the room
-  **5 Chat rooms** — General, Random, Gaming, Music, Coding
-  **Message history** — last 50 messages saved per room, loads when you join
-  **Typing indicator** — see when someone is typing
-  **Online user count** — live count of connected users
-  **Emoji picker** — 64 emojis to add to your messages
-  **Image sharing** — share images via URL
-  **Sound notifications** — ping when a new message arrives
-  **Private messaging** — click any username to open a DM panel
-  **Read receipts** — see when someone reads your message
-  **Custom username colors** — each user gets a unique color automatically
-  **Pin messages** — pin important messages to the top of the room
-  **Search messages** — search and highlight messages in real time
-  **Emoji reactions** — react to any message with 6 emojis
-  **Light / Dark mode** — toggle between themes with one click

Tech Stack

Tool	Purpose
Node.js	Server runtime
Express	HTTP server & static file serving
Socket.io	Real-time bidirectional communication
Vanilla HTML/CSS/JS Frontend — no framework needed	

Getting Started

Prerequisites

- [Node.js](#) v18 or higher
- npm (comes with Node.js)

Installation

Step 1 — Clone the repository

```
git clone https://github.com/YOUR_USERNAME/chat-app.git
cd chat-app
```

Step 2 — Install dependencies

```
npm install
```

Step 3 — Start the server

```
node server.js
```

Step 4 — Open in browser

Go to **http://localhost:3000** and start chatting!

Project Structure

```
chat-app/
├── public/
│   ├── index.html    ← Main chat UI
│   ├── style.css      ← All styling including light/dark themes
│   └── client.js      ← Frontend socket logic and UI interactions
├── server.js          ← Express + Socket.io server
└── package.json       ← Project dependencies
```

How to Use

Joining the Chat

1. Open **http://localhost:3000**

2. Type a nickname and press **Join Chat**
3. You'll land in the **General** room automatically

Switching Rooms

- Click any room in the left sidebar
- Message history loads automatically for each room

Private Messages (DM)

1. Click on any **username** in the chat
2. A DM panel pops up in the bottom right
3. Type and send — only you and the recipient can see it

Reactions

1. Hover over any message
2. Click 😊 **React**
3. Pick an emoji from the reaction picker

Pinning Messages

1. Hover over any message
2. Click 📌 **Pin**
3. The message appears in the pin bar at the top of the room

Searching Messages

- Type in the **search box** in the top right
- Matching messages get highlighted in real time

Image Sharing

1. Click the 🖼️ **button** next to the input
2. Paste any image URL starting with `http`
3. The image appears in chat for everyone

Light / Dark Mode

- Click the 🌙 **button** in the top right to toggle themes

Username Colors

Each user is automatically assigned a unique color from this palette:

Color	Hex
Red	#e94560
Green	#4ade80
Blue	#60a5fa
Amber	#f59e0b
Purple	#a78bfa
Teal	#34d399
Pink	#f472b6
Sky	#38bdf8

Socket.io Events

Client → Server

Event	Payload	Description
set nickname	nickname	Set user's display name
join room	room	Join a chat room
chat message	{room, message, image}	Send a message
typing	room	Notify others you're typing
stop typing	room	Stop typing notification
private message	{to, message}	Send a DM
pin message	{room, msg}	Pin a message
unpin message	room	Remove pinned message
react	{room, msgId, emoji}	React to a message
read receipt	{room, reader}	Notify message was read

Server → Client

Event	Payload	Description
chat message	message object	New message in room
message history	array of messages	Last 50 messages on join
user joined	{nickname, count}	Someone joined
user left	{nickname, count}	Someone left
typing	nickname	Someone is typing
stop typing	—	Someone stopped typing
private message	message object	Incoming DM
private message sent	message object	DM sent confirmation
pinned message	message object	Pin updated
reaction updated	{msgId, reactions}	Reactions updated
read receipt	reader	Message was read



Troubleshooting

Problem

Fix

Port 3000 already in use Run `lsof -i :3000` then `kill -9 PID`

Messages not appearing Check browser console for errors

DM not working Make sure both users are in the same server session

Sound not playing Click anywhere on the page first (browser autoplay policy)



Deployment

Deploy to Railway (Free)

1. Push your project to GitHub
2. Go to railway.app and sign up free
3. Click **New Project** → **Deploy from GitHub**
4. Select your repo → Railway auto-detects Node.js
5. Your app is live with a public URL! 🚀

Environment Variables

No environment variables needed for basic setup.



License

This project is open source and available under the [MIT License](#).

Built with ❤️ as a software engineering project. Real-time communication powered by Socket.io.